14、CAN tx/rx的回调和发送接收流程梳理

卖家'漩涡遨游aa' 专属资料，非在漩涡遨游aa 购买资料，不提供答疑(一次购买仅提供有限次数10次答疑)

一、CAN 报文发送

https://blog.csdn.net/m0_58346868/article/details/124897133

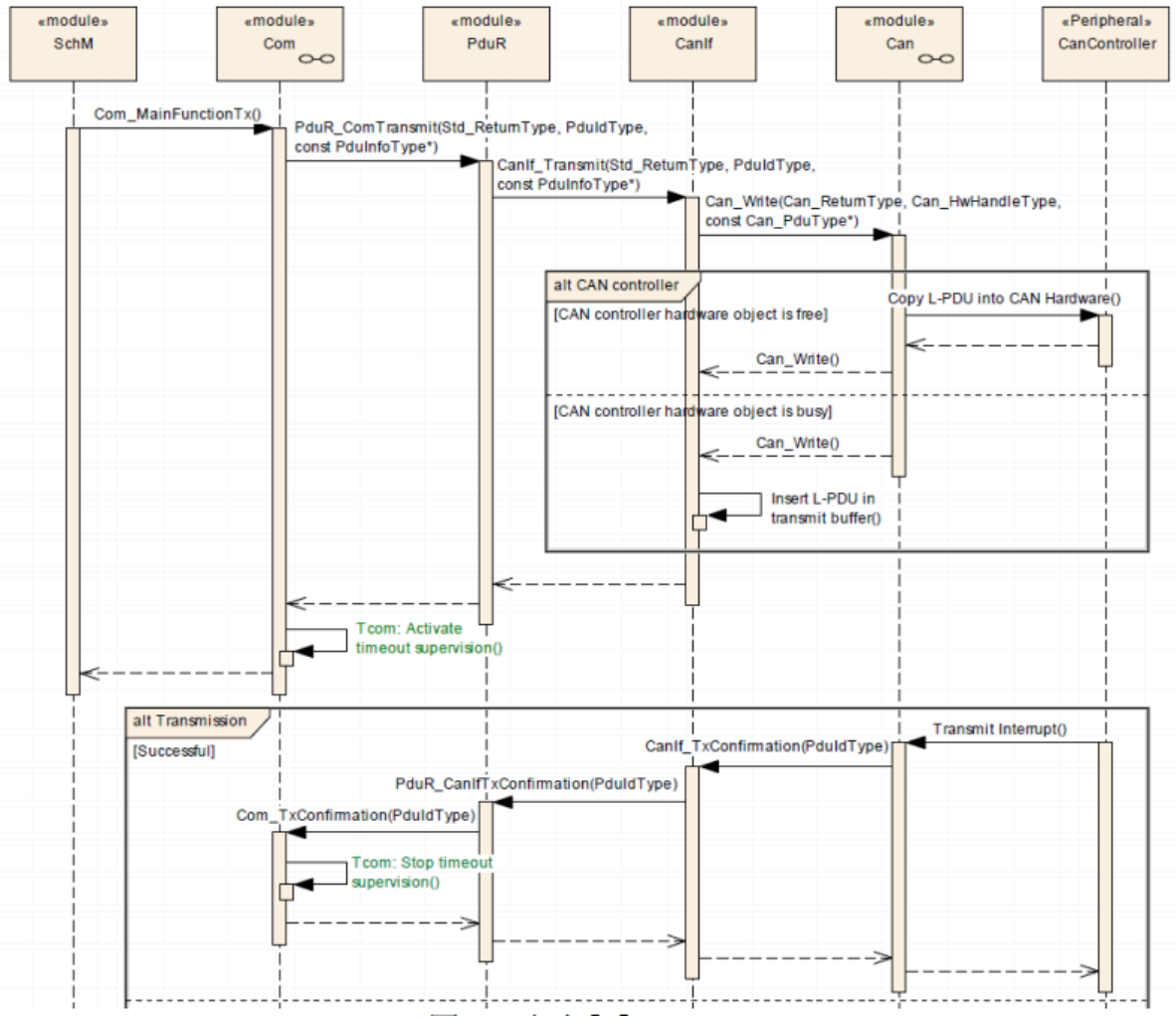
https://blog.csdn.net/weixin_45798899/article/details/135236900

https://blog.csdn.net/NMR0574/article/details/129660493?utm_medium=distribute.pc_relevant.none-task-blog-2~default~baidujs_baidulandingword~default-0-129660493-blog-131826499.235^v43^pc_blog_bottom_relevance_base7&spm=1001.2101.3001.4242.1&utm_relevant_index=3

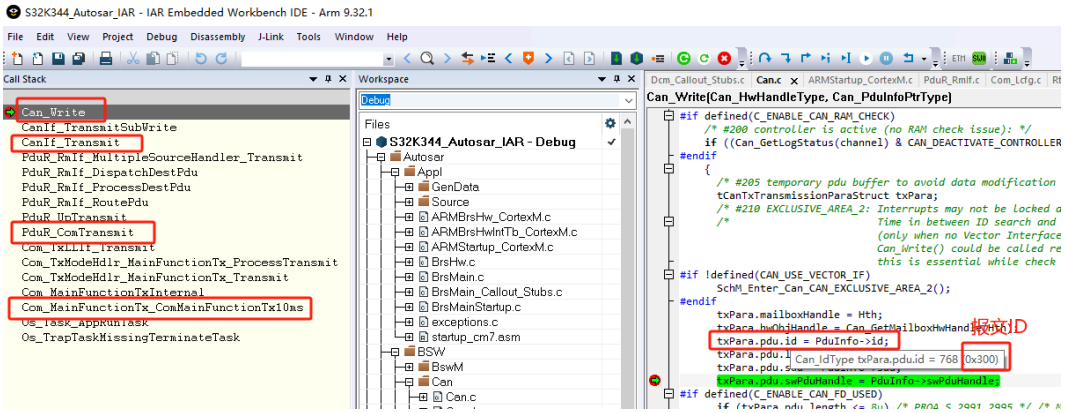
https://blog.csdn.net/tjcwt2011/article/details/136515463

1.1、周期报文发送流程

整体发送流程和发送确认流程如下

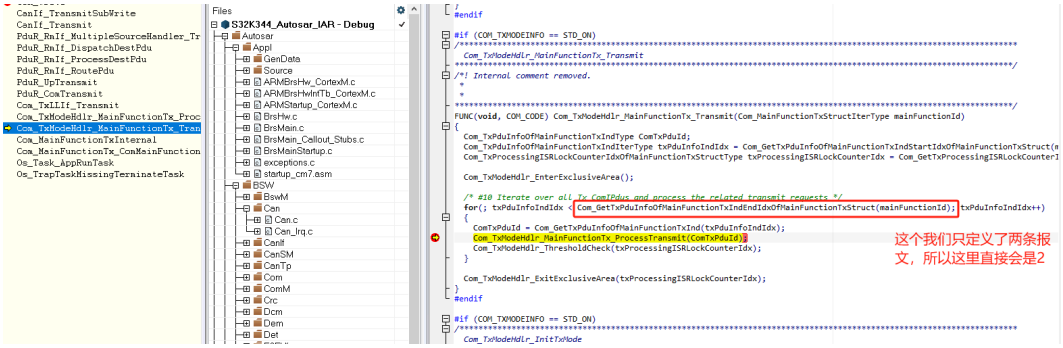


Can发送，报文ID在txPara.pdu.id = PduInfo→id; 长度在txPara.pdu.length = PduInfo→length;

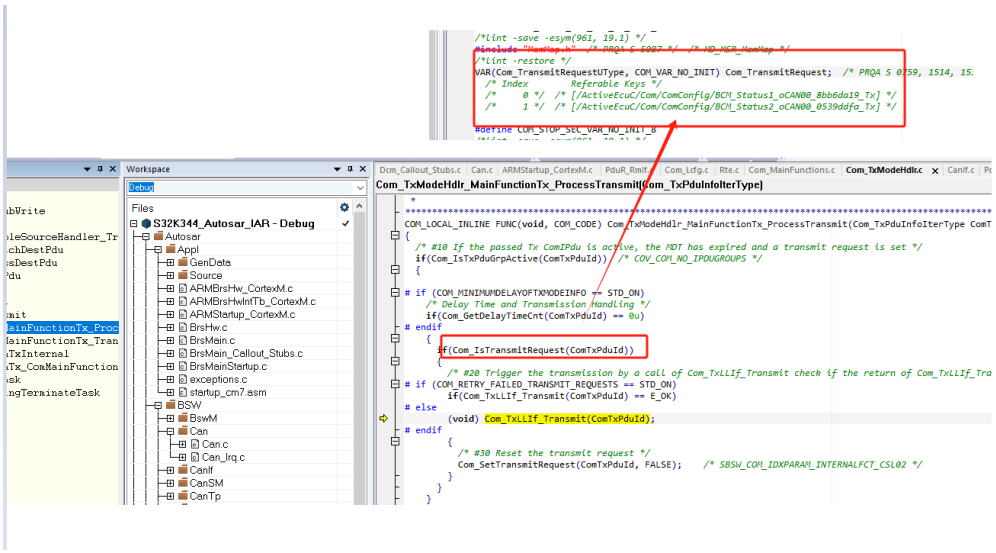


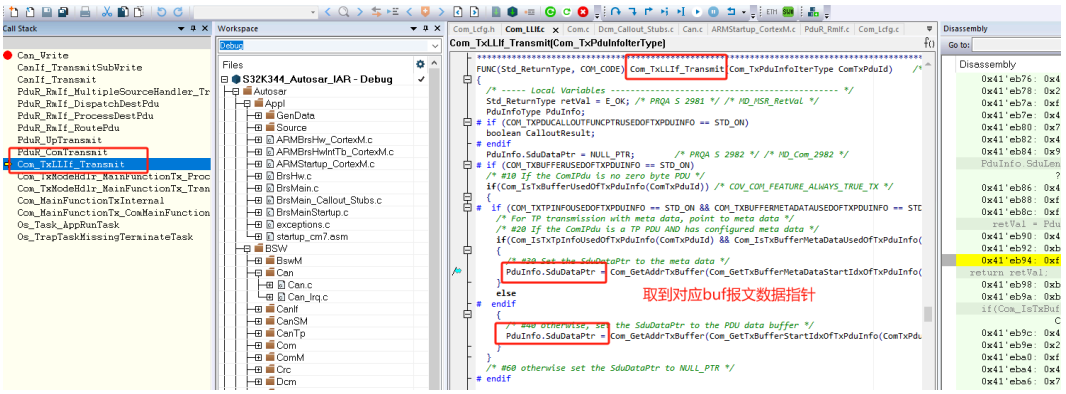
com

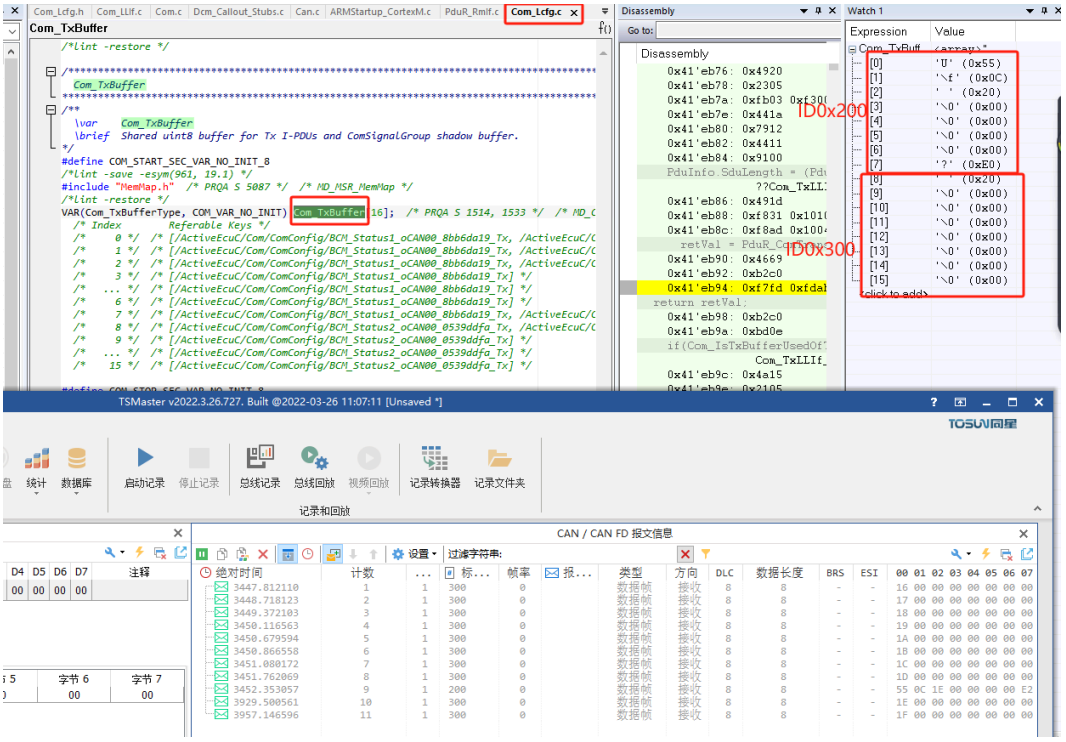
这里ID就是0和1



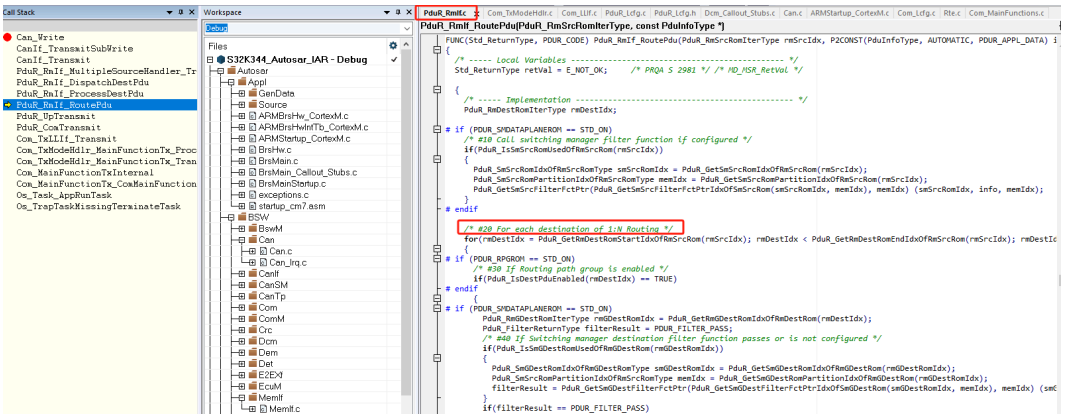
#define Com_GetTxPduInfoOfMainFunctionTxIndEndIdxOfMainFunctionTxStruct(Index) ((Com_TxPduInfoOfMainFunctionTxIndEndIdxOfMainFunctionTxStructType) ((((Com_TxPduInfoOfMainFunctionTxIndEndIdxOfMainFunctionTxStructType)(Index)) + 2u))) /**< the end index of the 0:n relation pointing to Com_TxPduInfoOfMainFunctionTxInd */

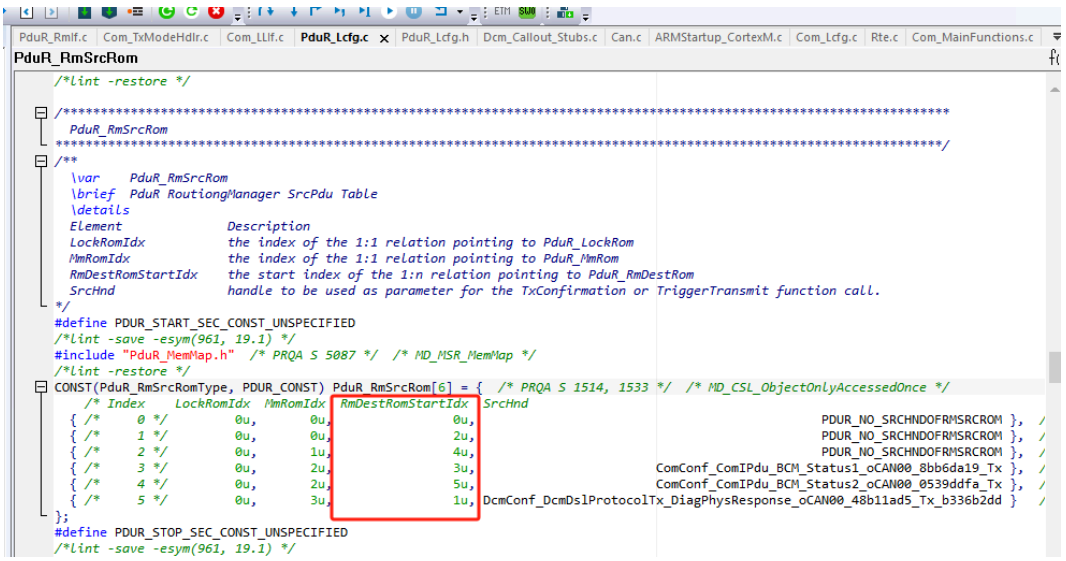





Com_TxBuffer
ID0x200
ID0x300
TSMaster v2022.3.26.727. Built @2022-03-26 11:07:11 [Unsaved *]
CAN / CAN FD 报文信息

路由




PduR_RmSrcRom
/*lint -restore */
/**
\var PduR_RmSrcRom
\brief PduR RoutingManager SrcPdu Table
\details
Element Description
LockRomIdx the index of the 1:1 relation pointing to PduR_LockRom
MmRomIdx the index of the 1:1 relation pointing to PduR_MmRom
RmDestRomStartIdx the start index of the 1:n relation pointing to PduR_RmDestRom
SrcHnd handle to be used as parameter for the TxConfirmation or TriggerTransmit function call.
*/
#define PDUR_START_SEC_CONST_UNSPECIFIED
/*lint -save -esym(961, 19.1) */
#include "PduR_MemMap.h" /* PRQA S 5087 */ /* MD_MSR_MemMap */
/*lint -restore */
CONST(PduR_RmSrcRomType, PDUR_CONST) PduR_RmSrcRom[6] = { /* PRQA S 1514, 1533 */ /* MD_CSL_ObjectOnlyAccessedOnce */
/* Index LockRomIdx MmRomIdx RmDestRomStartIdx SrcHnd
{ /* 0 */ 0u, 0u, 0u, PDUR_NO_SRCHNDOFRMSRCROM },
{ /* 1 */ 0u, 0u, 2u, PDUR_NO_SRCHNDOFRMSRCROM },
{ /* 2 */ 0u, 1u, 4u, PDUR_NO_SRCHNDOFRMSRCROM },
{ /* 3 */ 0u, 2u, 3u, ComConf_ComIPdu_BCM_Status1_oCAN00_8bb6da19_Tx },
{ /* 4 */ 0u, 2u, 5u, ComConf_ComIPdu_BCM_Status2_oCAN00_0539ddfa_Tx },
{ /* 5 */ 0u, 3u, 1u, DcmConf_DcmDslProtocolTx_DiagPhysResponse_oCAN00_48b11ad5_Tx_b336b2dd }
};
#define PDUR_STOP_SEC_CONST_UNSPECIFIED
/*lint -save -esym(961, 19.1) */

RmDestRomStartIdx 和 rmDestIdx 对应

```
PduR_RmIf_ProcessDestPdu(PduR_RmDestRomIterType, const PduInfoType *)
  *
  *
  ***********************************************************************************************************
 FUNC(Std_ReturnType, PDUR_CODE) PduR_RmIf_ProcessDestPdu(PduR_RmDestRomIterType rmDestIdx, P2CONST(PduInfoType, AUTOM
 {
   Std_ReturnType retVal = E_NOT_OK;     /* PRQA S 2981 */ /* MD_MSR_RetVal */

   switch (PduR_GetPduLengthHandlingStrategyOfRmDestRom(rmDestIdx))
   {
 # if(PDUR_EXISTS_SHORTEN_PDULENGTHHANDLINGSTRATEGYOFRMDESTROM == STD_ON)
     case PDUR_SHORTEN_PDULENGTHHANDLINGSTRATEGYOFRMDESTROM:
       {
         if (PduR_RmIf_CheckInfoPointer(info, PDUR_FCT_RMIF_ROUTE) == E_OK)  /* SBSW_PDUR_VALID_PTR_ENSURED_BY_CALLING_
         {
           PduInfoType destPduInfo;
           destPduInfo.SduDataPtr = info->SduDataPtr;
           destPduInfo.SduLength = PduR_RmIf_TruncateDlc(PduR_GetRmGDestRomIdxOfRmDestRom(rmDestIdx), info);  /* SBSW_
           retVal = PduR_RmIf_DispatchDestPdu(rmDestIdx, &destPduInfo);  /* SBSW_PDUR_VALID_PTR_ENSURED_BY_CALLING_FUN
         }
       }
       break;
 # endif
 # if(PDUR_EXISTS_IGNORE_PDULENGTHHANDLINGSTRATEGYOFRMDESTROM == STD_ON)
     case PDUR_IGNORE_PDULENGTHHANDLINGSTRATEGYOFRMDESTROM:
       {
         retVal = PduR_RmIf_DispatchDestPdu(rmDestIdx, info);  /* SBSW_PDUR_VALID_PTR_ENSURED_BY_CALLING_FUNCTION */
       }
       break;
 # endif
 # if(PDUR_EXISTS_DISCARD_PDULENGTHHANDLINGSTRATEGYOFRMDESTROM == STD_ON)
     case PDUR_DISCARD_PDULENGTHHANDLINGSTRATEGYOFRMDESTROM:
       if (PduR_RmIf_CheckInfoPointer(info, PDUR_FCT_RMIF_ROUTE) == E_OK)  /* SBSW_PDUR_VALID_PTR_ENSURED_BY_CALLING_F
       {
         if (info->SduLength <= PduR_GetMaxPduLengthOfRmGDestRom(PduR_GetRmGDestRomIdxOfRmDestRom(rmDestIdx)))
         {
           retVal = PduR_RmIf_DispatchDestPdu(rmDestIdx, info);  /* SBSW_PDUR_VALID_PTR_ENSURED_BY_CALLING_FUNCTION */
         }
       }
```

1.2、报文周期关键信息

报文周期决定

下面这个变量是代表com tx时报文是否到发送的时间

```
Com_Lcfg.h | Com_TxModeHdlr.h | Dcm_Callout_Stubs.c | Can.c | ARMStartup_CortexM.c | PduR_Rmlf.c | Com_Lcfg.c | Rte.c | Com_MainFunctions.c | Com_TxModeHdlr.c
Com_GetSizeOfRxPduInfoOfMainFunctionRxIndOfPCConfig()          2u  /**< the number of accomplish
Com_GetSizeOfRxPduInfoOfPCConfig()                             2u  /**< the number of accomplish
Com_GetSizeOfRxSigBufferSInt8OfPCConfig()                      2u  /**< the number of accomplish
Com_GetSizeOfRxSigBufferUInt16OfPCConfig()                     1u  /**< the number of accomplish
Com_GetSizeOfRxSigInfoOfPCConfig()                             3u  /**< the number of accomplish
Com_GetSizeOfTxAccessInfoOfPCConfig()                          4u  /**< the number of accomplish
Com_GetSizeOfTxBufferOfPCConfig()                              16u  /**< the number of accomplis
Com_GetSizeOfTxCyclicPduOfMainFunctionTxIndOfPCConfig()        2u  /**< the number of accomplish
Com_GetSizeOfTxCyclicPduOfPCConfig()                           2u  /**< the number of accomplish
Com_GetSizeOfTxIPduGroupISRLockCounterOfPCConfig()             1u  /**< the number of accomplish
Com_GetSizeOfTxModeFalseOfPCConfig()                           2u  /**< the number of accomplish
Com_GetSizeOfTxModeInfoOfPCConfig()                            2u  /**< the number of accomplish
Com_GetSizeOfTxModeTrueOfPCConfig()                            2u  /**< the number of accomplish
Com_GetSizeOfTxPduInfoIndOfPCConfig()                          2u  /**< the number of accomplish
Com_GetSizeOfTxPduInfoOfMainFunctionTxIndOfPCConfig()          2u  /**< the number of accomplish
Com_GetSizeOfTxPduInfoOfPCConfig()                             2u  /**< the number of accomplish
Com_GetSizeOfTxPduInitValueOfPCConfig()                        16u  /**< the number of accomplis
Com_GetSizeOfTxProcessingISRLockCounterOfPCConfig()            1u  /**< the number of accomplish
Com_GetSizeOfTxSigInfoOfPCConfig()                             4u  /**< the number of accomplish
Com_GetTransmitRequestOfPCConfig()                             Com_TransmitRequest.raw  /**< the
Com_GetTxAccessInfoOfPCConfig()                                Com_TxAccessInfo  /**< the pointe
Com_GetTxBufferOfPCConfig()                                    Com_TxBuffer  /**< the pointer to
Com_GetTxCyclicPduOfMainFunctionTxIndOfPCConfig()              Com_TxCyclicPduOfMainFunctionTxIn
Com_GetTxCyclicPduOfPCConfig()                                 Com_TxCyclicPdu  /**< the pointer
Com_GetTxIPduGroupISRLockCounterOfPCConfig()                   Com_TxIPduGroupISRLockCounter.raw
Com_GetTxIPduGroupISRLockThresholdOfPCConfig()                 1u
```

```
Com_MainFunctionTxInternal(Com_MainFunctionTxStructIterType)
FUNC(void, COM_CODE) Com_MainFunctionTxInternal(Com_MainFunctionTxStructIterType mainFunctionId)
{
/* #10 Check if COM is initialized */
  if(Com_IsInitialized())
  {
    /* #20 check that id is smaller then the size of the MainFunctionTxStruct */
    if(mainFunctionId < Com_GetSizeOfMainFunctionTxStruct())    /* COV_COM_SILENT_TX */
    {
# if ((COM_PERIODICOFTXMODETRUE == STD_ON) || (COM_PERIODICOFTXMODEFALSE == STD_ON) || (COM_REPCNTOFTXMODETRUE == STD_O
      /* #30 Do TxMode processing */
      Com_TxModeHdlr_MainFunctionTx_CycleCounter(mainFunctionId);
# endif

# if (COM_TXTOUTINFO == STD_ON)
      /* #40 Do Tx deadline monitoring, if configured */
      Com_TxDlMon_MainFunctionTx(mainFunctionId);
# endif

# if (COM_GWTIMEOUTINFO == STD_ON)
      /* #50 Handle GwTimeout, if configured */
      Com_GwTout_HandleCounter(mainFunctionId);
# endif

      /* #60 Process transmit request and initiate transition */
      Com_TxModeHdlr_MainFunctionTx_Transmit(mainFunctionId);

# if (COM_CBKTXACKDEFFUNCPTRINDUSEDOFTXPDUINFO == STD_ON)
      /* #70 Process deferred Tx Confirmations */
      Com_TxNHdlr_MainFunctionTx(mainFunctionId);
# endif
    }
  }
```

里面会做cnt--，到计时后置raw的标志位

判断是否时周期发送的请求报文

每次Cnt计数到0时重新加载新的值

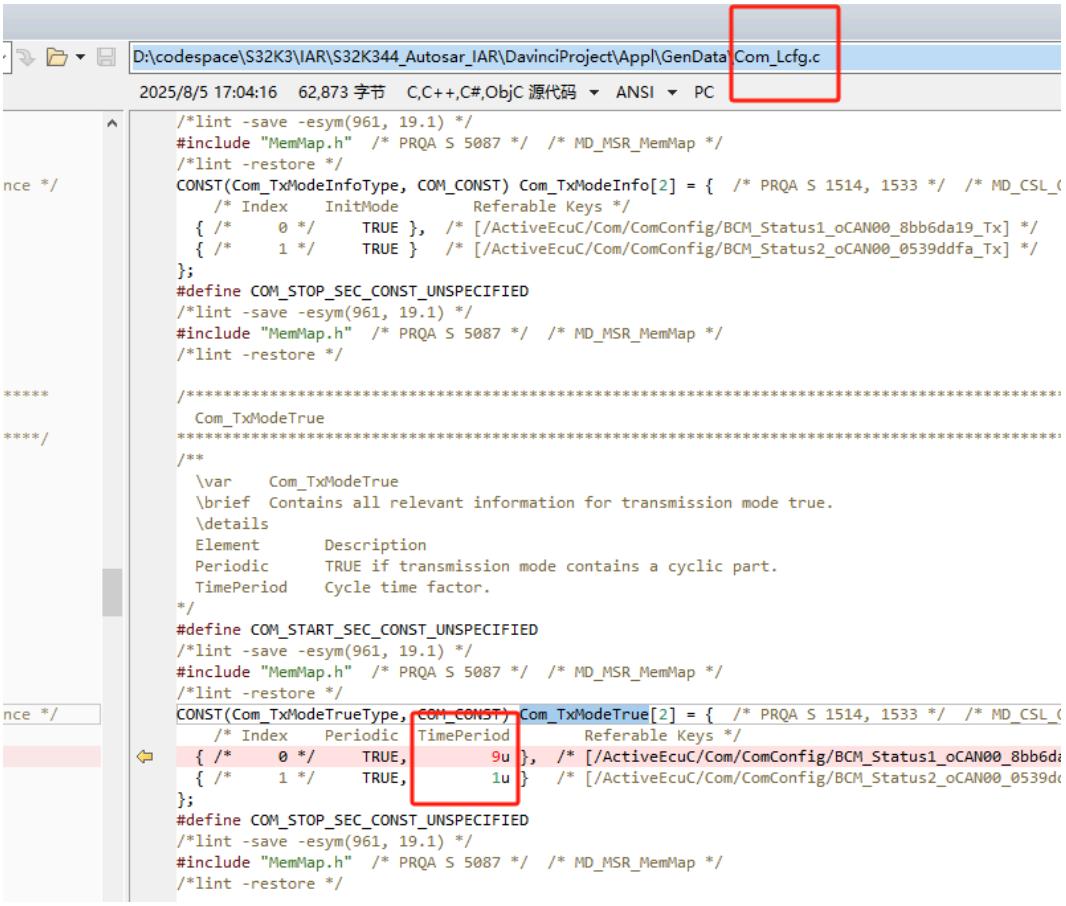
设raw标志位

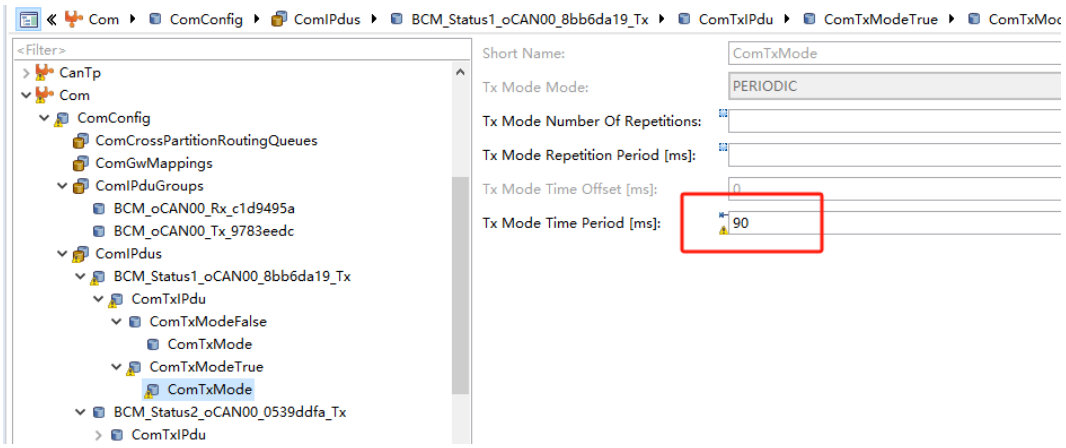
重新加载计数值

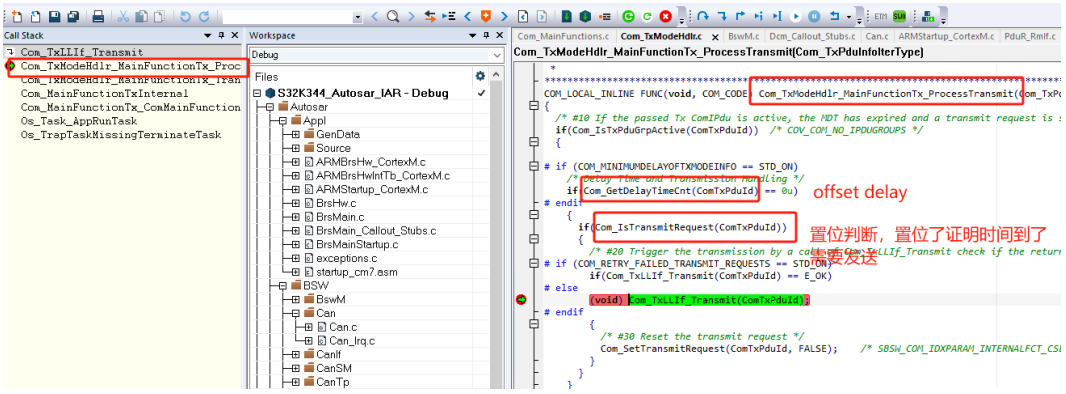
Com_GetCycleTimeCntOfPCConfig() 是cnt的全局变量

```
Com_GetConfigIdOfPCConfig()                                  0u  /**< DefinitionRef: /MICF
Com_GetConstValueSInt8OfPCConfig()                           Com_ConstValueSInt8  /**< the
Com_GetConstValueUInt16OfPCConfig()                          Com_ConstValueUInt16  /**< th
Com_GetCurrentTxModeOfPCConfig()                             Com_CurrentTxMode.raw  /**< t
Com_GetCycleTimeCntOfPCConfig()                              Com_CycleTimeCnt.raw  /**< th
Com_GetCyclicSendRequestOfPCConfig()                         Com_CyclicSendRequest.raw  /*
Com_GetHandleRxPduDeferredOfPCConfig()                       Com_HandleRxPduDeferred.raw
Com_GetIPduGroupInfoOfPCConfig()                             Com_IPduGroupInfo  /**< the p
```

计数的加载值来源于ComTxModeTrue，我这里修改为9，对应报文就是90ms周期


D:\codespace\S32K3\IAR\S32K344_Autosar_IAR\DavinciProject\Appl\GenData\Com_Lcfg.c
2025/8/5 17:04:16 62,873 字节 C,C++,C#,ObjC 源代码 ANSI PC
/*lint -save -esym(961, 19.1) */
#include "MemMap.h" /* PRQA S 5087 */ /* MD_MSR_MemMap */
/*lint -restore */
CONST(Com_TxModeInfoType, COM_CONST) Com_TxModeInfo[2] = { /* PRQA S 1514, 1533 */ /* MD_CSL_
/* Index InitMode Referable Keys */
{ /* 0 */ TRUE }, /* [/ActiveEcuC/Com/ComConfig/BCM_Status1_oCAN00_8bb6da19_Tx] */
{ /* 1 */ TRUE } /* [/ActiveEcuC/Com/ComConfig/BCM_Status2_oCAN00_0539ddfa_Tx] */
};
#define COM_STOP_SEC_CONST_UNSPECIFIED
/*lint -save -esym(961, 19.1) */
#include "MemMap.h" /* PRQA S 5087 */ /* MD_MSR_MemMap */
/*lint -restore */
Com_TxModeTrue
/**
\var Com_TxModeTrue
\brief Contains all relevant information for transmission mode true.
\details
Element Description
Periodic TRUE if transmission mode contains a cyclic part.
TimePeriod Cycle time factor.
*/
#define COM_START_SEC_CONST_UNSPECIFIED
/*lint -save -esym(961, 19.1) */
#include "MemMap.h" /* PRQA S 5087 */ /* MD_MSR_MemMap */
/*lint -restore */
CONST(Com_TxModeTrueType, COM_CONST) Com_TxModeTrue[2] = { /* PRQA S 1514, 1533 */ /* MD_CSL_
/* Index Periodic TimePeriod Referable Keys */
{ /* 0 */ TRUE, 9u }, /* [/ActiveEcuC/Com/ComConfig/BCM_Status1_oCAN00_8bb6da
{ /* 1 */ TRUE, 1u } /* [/ActiveEcuC/Com/ComConfig/BCM_Status2_oCAN00_0539d
};
#define COM_STOP_SEC_CONST_UNSPECIFIED
/*lint -save -esym(961, 19.1) */
#include "MemMap.h" /* PRQA S 5087 */ /* MD_MSR_MemMap */
/*lint -restore */

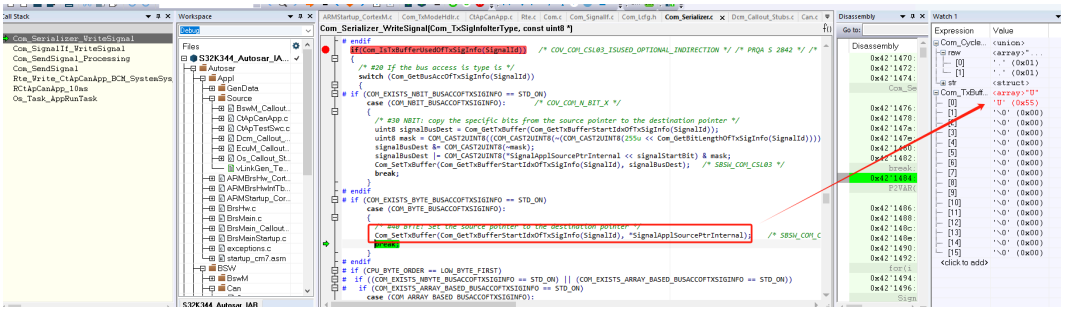

Com ▸ ComConfig ▸ ComIPdus ▸ BCM_Status1_oCAN00_8bb6da19_Tx ▸ ComTxIPdu ▸ ComTxModeTrue ▸ ComTxMod
<Filter>
CanTp
Com
ComConfig
ComCrossPartitionRoutingQueues
ComGwMappings
ComIPduGroups
BCM_oCAN00_Rx_c1d9495a
BCM_oCAN00_Tx_9783eedc
ComIPdus
BCM_Status1_oCAN00_8bb6da19_Tx
ComTxIPdu
ComTxModeFalse
ComTxMode
ComTxModeTrue
ComTxMode
BCM_Status2_oCAN00_0539ddfa_Tx
ComTxIPdu
Short Name: ComTxMode
Tx Mode Mode: PERIODIC
Tx Mode Number Of Repetitions:
Tx Mode Repetition Period [ms]:
Tx Mode Time Offset [ms]: 0
Tx Mode Time Period [ms]: 90


Call Stack
Com_TxLLIf_Transmit
Com_TxModeHdlr_MainFunctionTx_Proc
Com_MainFunctionTxInternal
Com_MainFunctionTx_ComMainFunction
Os_Task_AppRunTask
Os_TrapTaskMissingTerminateTask
Workspace
Debug
Files
S32K344_Autosar_IAR - Debug
Com_TxModeHdlr_MainFunctionTx_ProcessTransmit(Com_TxPduInfoIterType)
COM_LOCAL_INLINE FUNC(void, COM_CODE) Com_TxModeHdlr_MainFunctionTx_ProcessTransmit(Com_TxP
/* #10 If the passed Tx ComIPdu is active, the MDT has expired and a transmit request is
if(Com_IsTxPduGrpActive(ComTxPduId)) /* COV_COM_NO_IPDUGROUPS */
# if (COM_MINIMUMDELAYOFTXMODEINFO == STD_ON)
if(Com_GetDelayTimeCnt(ComTxPduId) == 0u)
offset delay
# endif
if(Com_IsTransmitRequest(ComTxPduId))
置位判断，置位了证明时间到了
# if (COM_RETRY_FAILED_TRANSMIT_REQUESTS == STD_ON)
if(Com_TxLLIf_Transmit(ComTxPduId) == E_OK)
# else
(void) Com_TxLLIf_Transmit(ComTxPduId);
endif
/* #30 Reset the transmit request */
Com_SetTransmitRequest(ComTxPduId, FALSE);

Com_Init cnt都置为1，所以Com_MainFunctionTx_ComMainFunctionTx10ms 第一次调用时就会发送

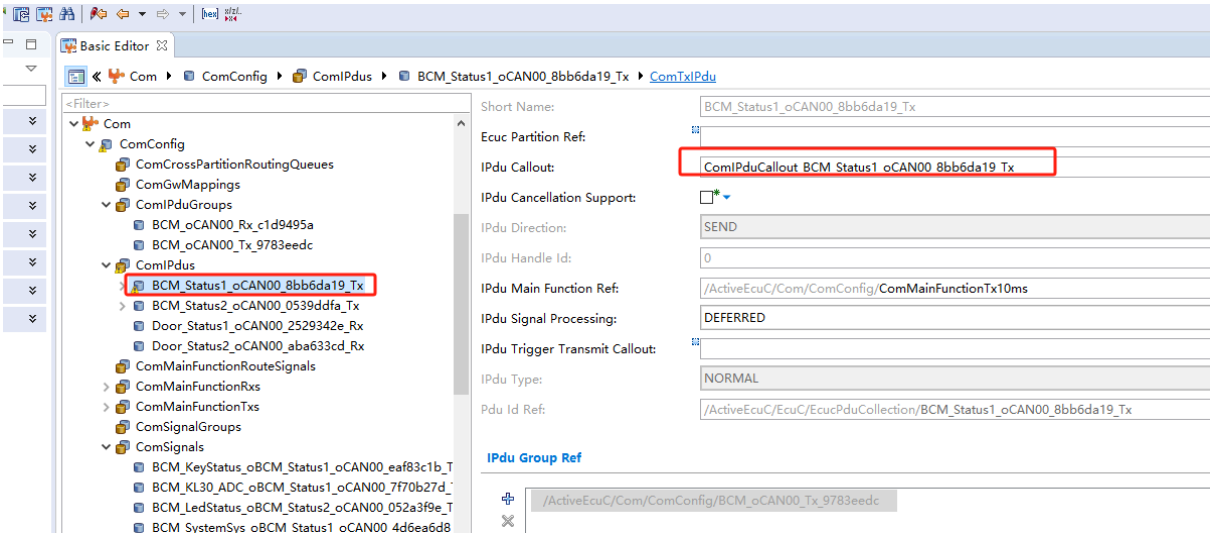
1.3、Com_SendSignal流程

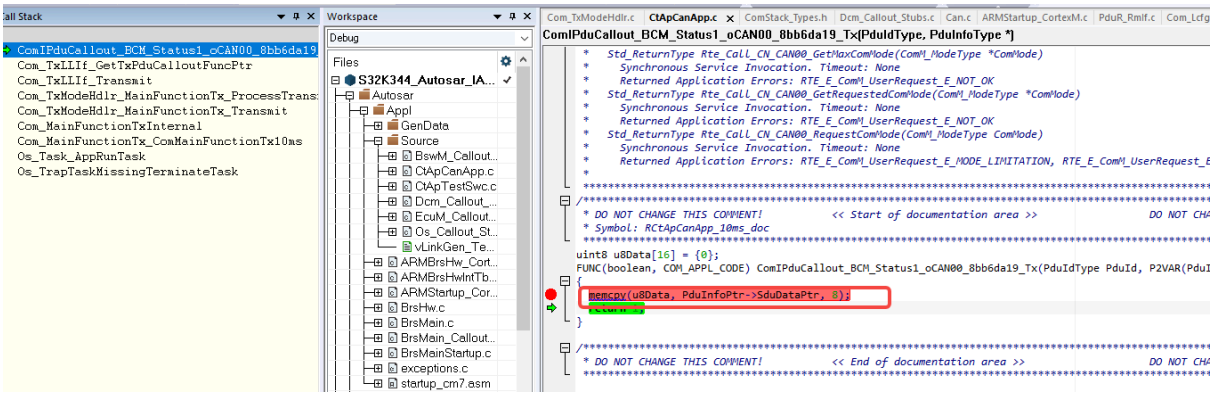
Com_SendSignal， 根据nbit 或者nbyte，将数据写入Com_TxBuffer对应的位置，至此Com_SendSignal工作完成，等待Com_MainFunction10msTx去发送数据



1.4、发送前回调 Tx ComIPduCallout

是在Com→ PduR前调用的用户回调接口，是数据发送前的回调，PduInfoPtr→SduDataPtr ，PduInfoPtr->SduLength 发送数据的指针和长度，返回TRUE 1, 代表回调成功





COM 发送处理

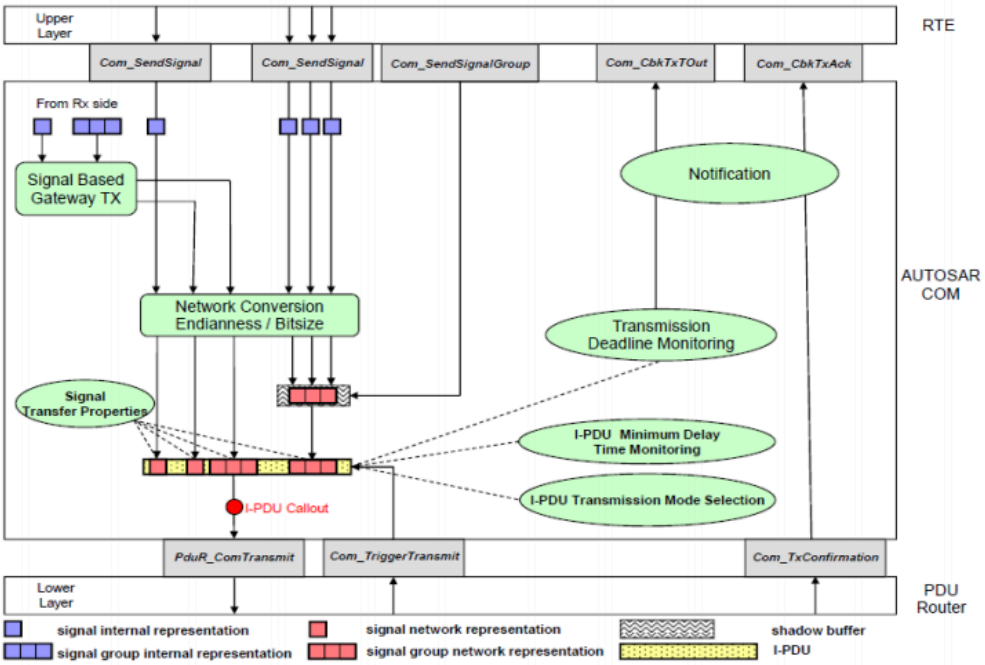
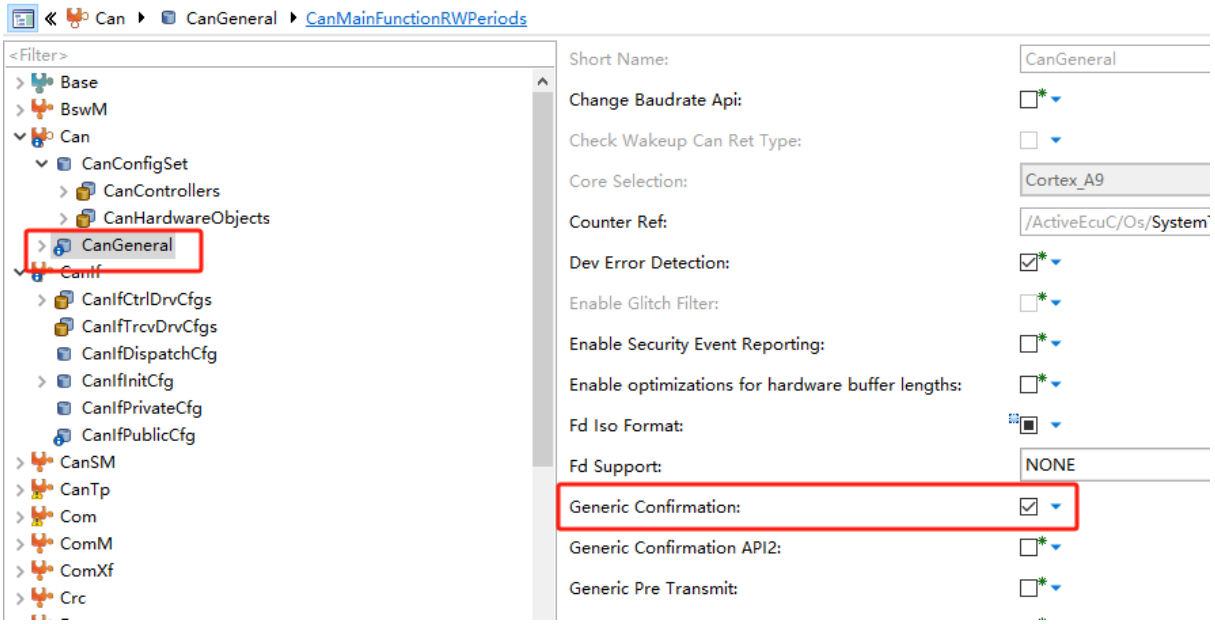


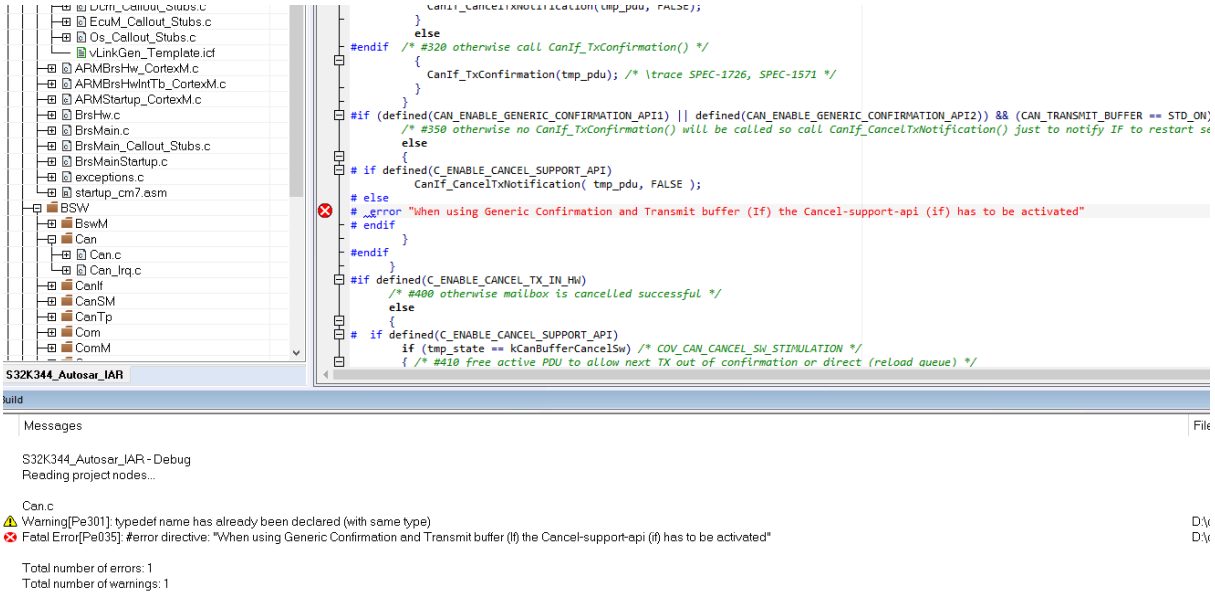
Figure 5 The AUTOSAR COM module's interaction model for transmission

1.5、发送确认回调 Appl_GenericConfirmation

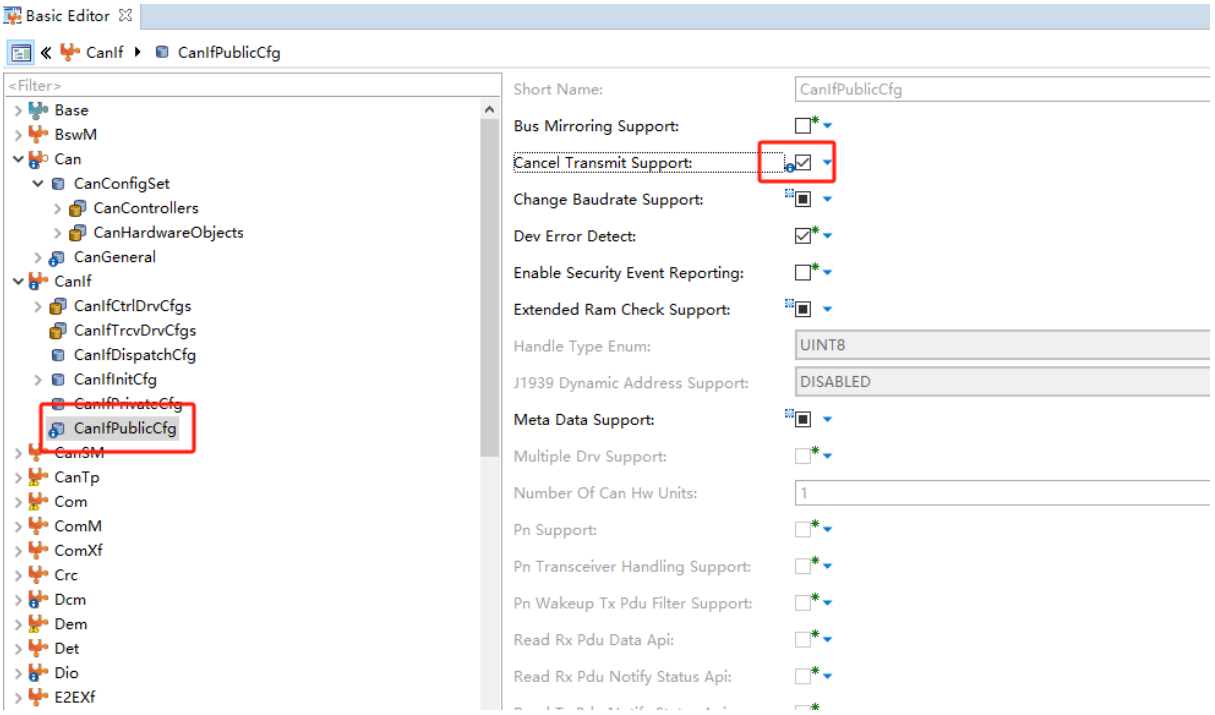
勾选下面，那么发送了报文并且接收节点接收完成后，can底层中断回调出来告知用户对方接收了



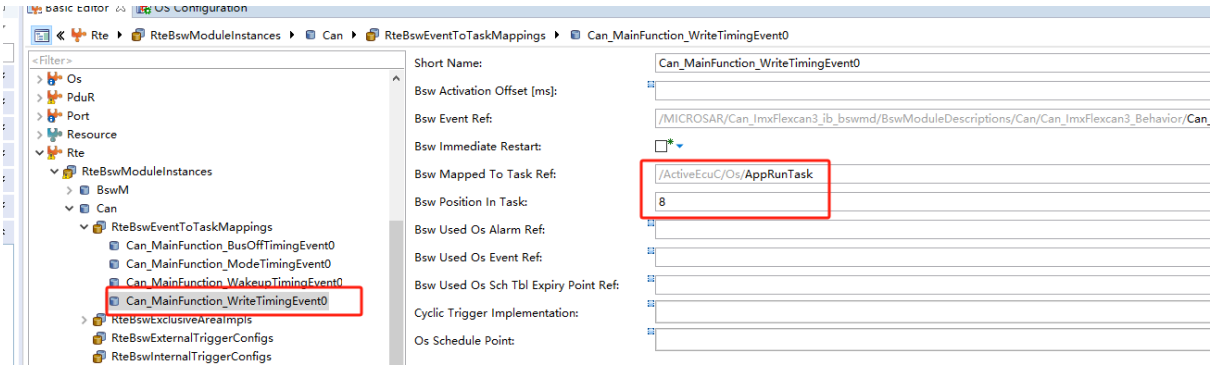
编译报错



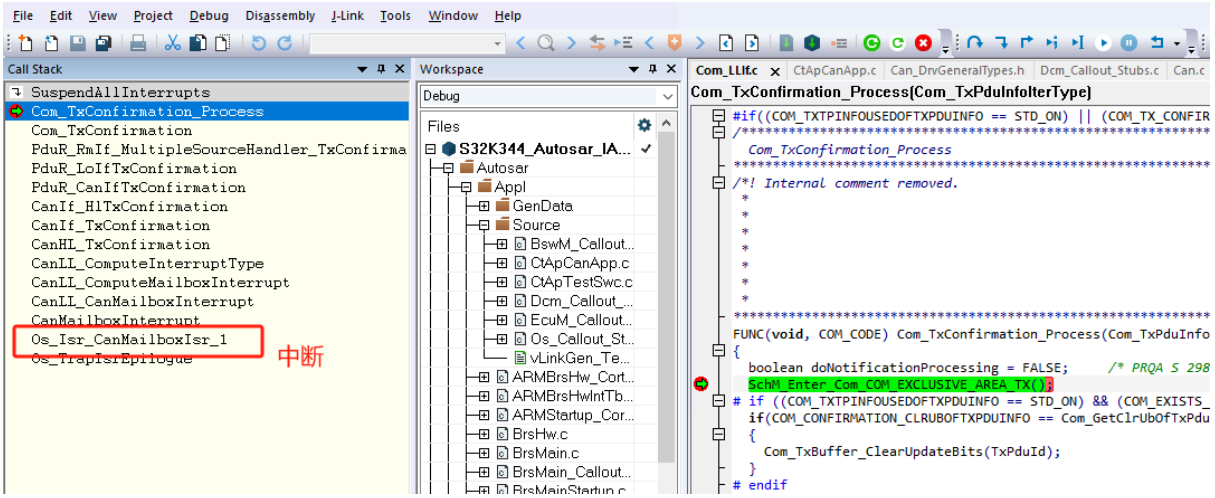
使能下面的勾选， CanIf 的cancel transmit ，然后重新生成Can/CanIf



Can_MainFunction_Write map到task



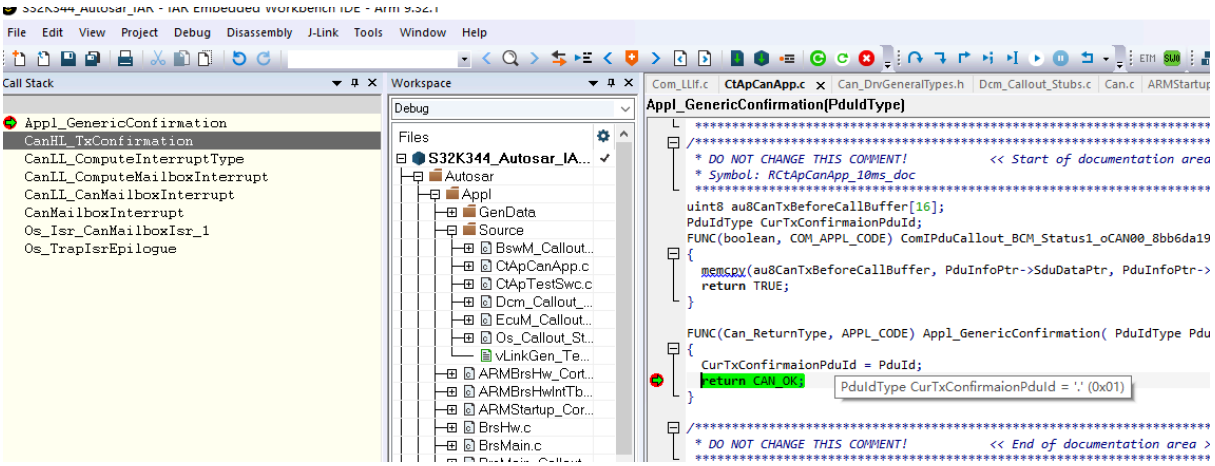
正常回调，从can中断→Canif→PduR→Com ， 一层层向上，当勾选Generic Confirmation 后，用户需要自己实现函数Appl_GenericConfirmation



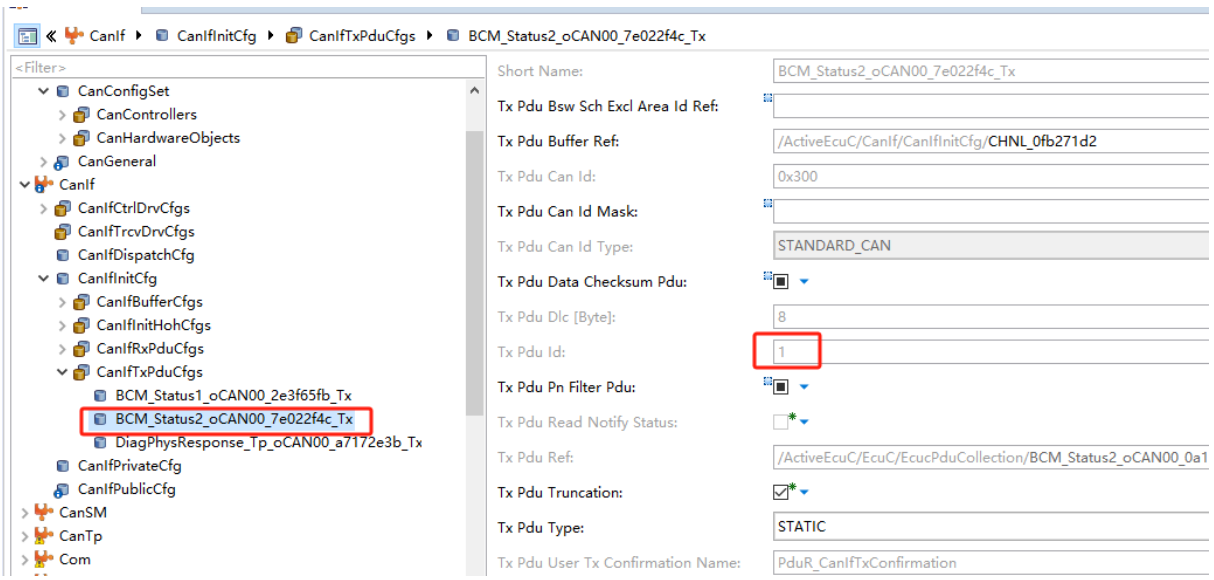
添加回调之后 在CanHl_TxConfirmation 中将会调用 用户的回调函数

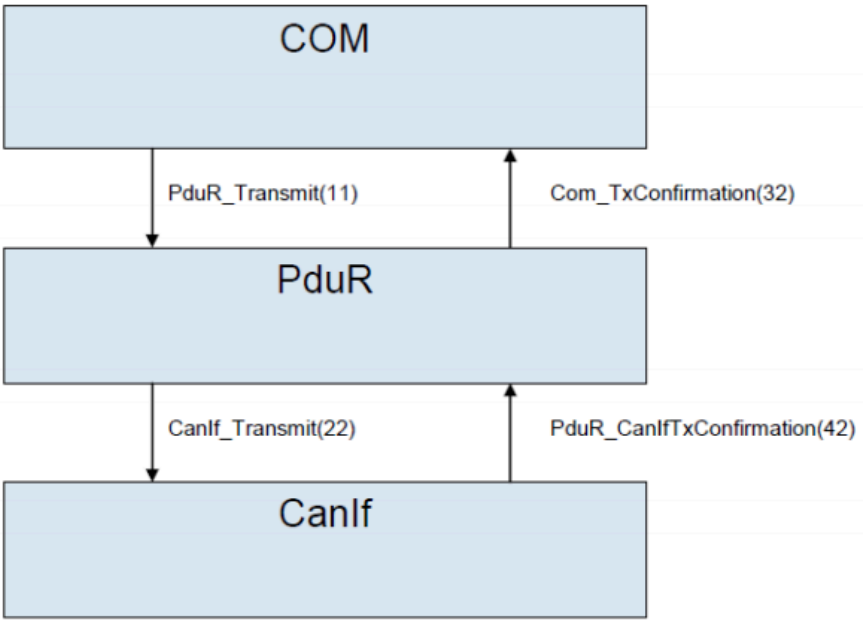
```
FUNC(Can_ReturnType, APPL_CODE) Appl_GenericConfirmation( PduIdType PduId )
{

}
```



这里回调传进来的参数跟PduId 跟CanIf 里面的Tx Pdu Id是对应的





二、报文接收

2.1、报文接收

COM 接收处理：

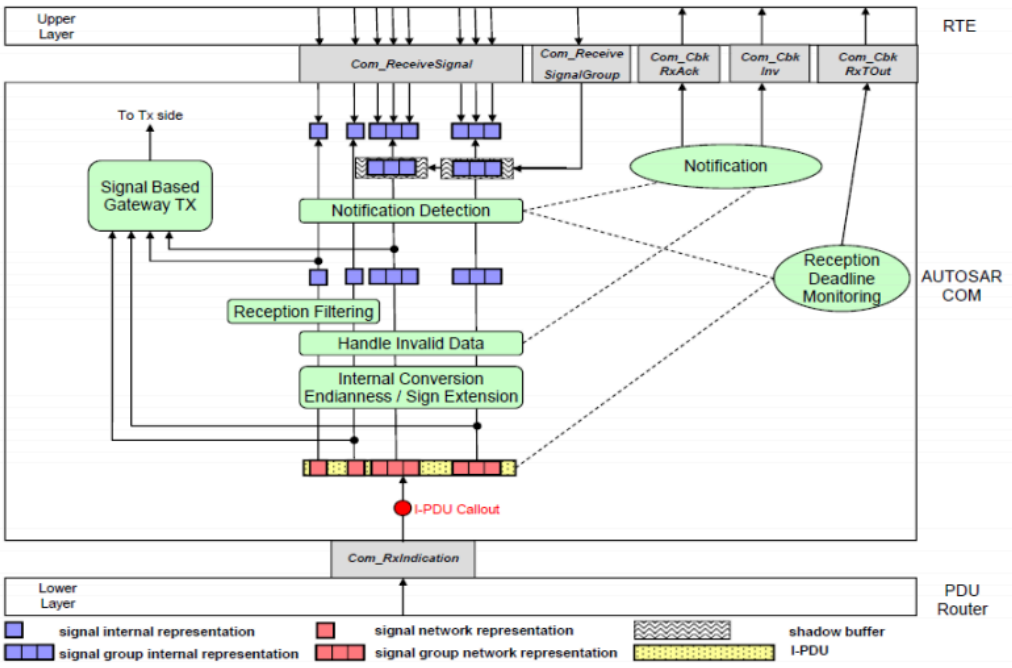
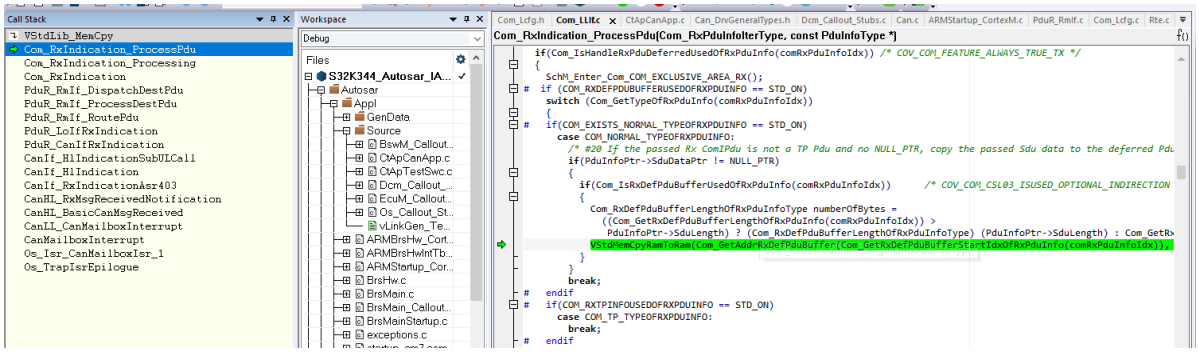
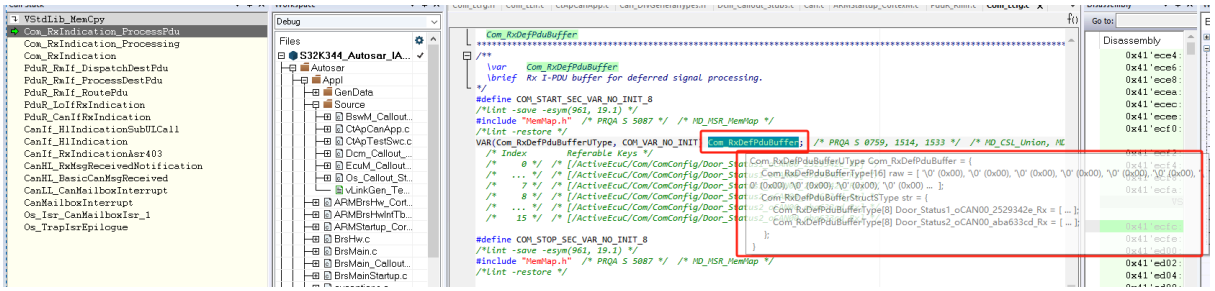


Figure 6 The AUTOSAR COM module's interaction model for reception

我们配置的是中断处理，接收到报文后，从中断层层递进调用, Can→CanIf RxIndication→PduR RxIndication→Com RxIndication

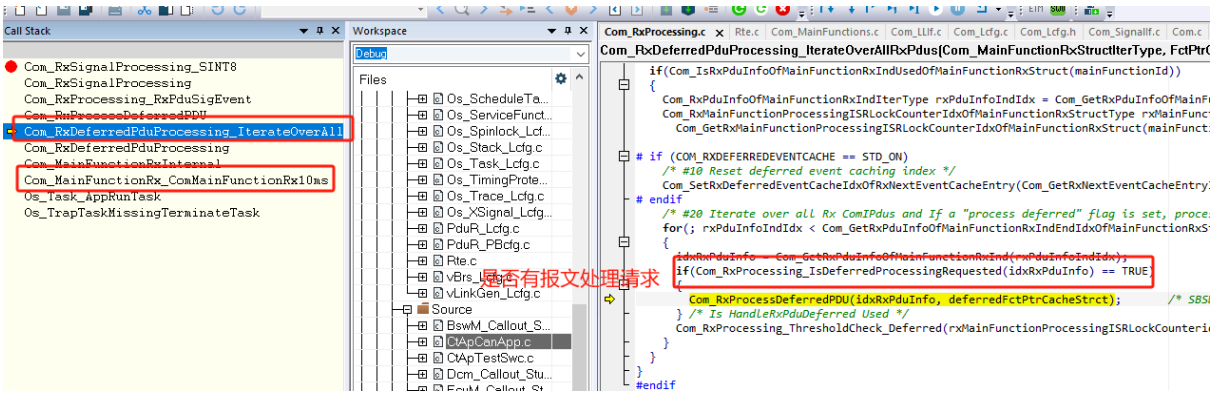
最终在下面的Memcpy 将数据拷贝到 Com_RxDefPduBuffer

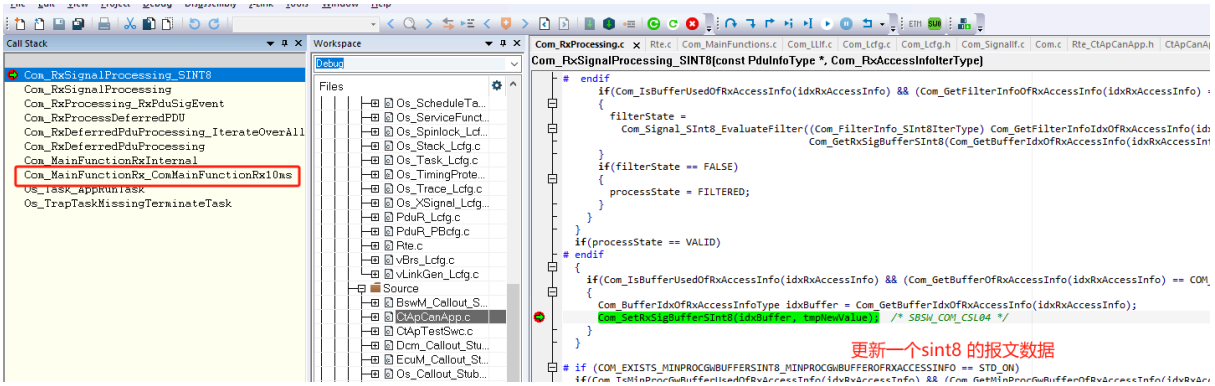


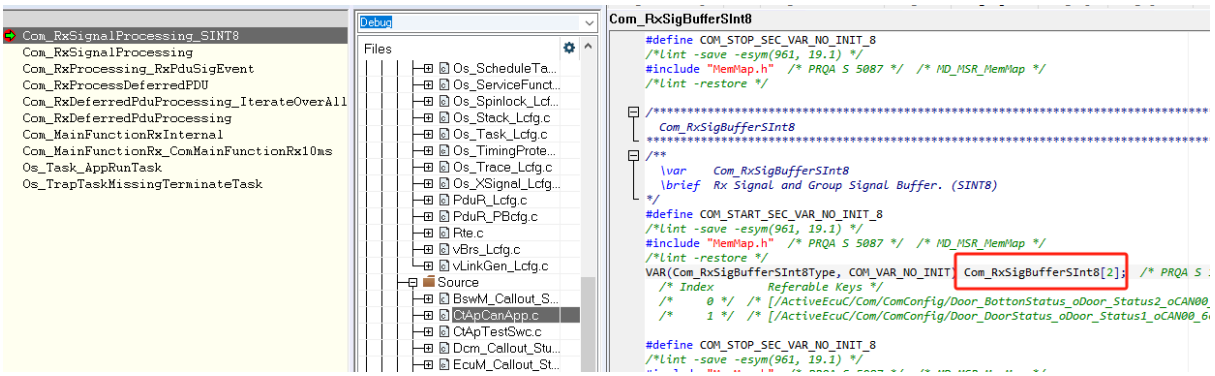


中断部分的工作完成

然后等Com_MainFunctionRx_ComMainFunctionRx10ms 去更新 Com_ReceiveSignal 中的各个信号buffer



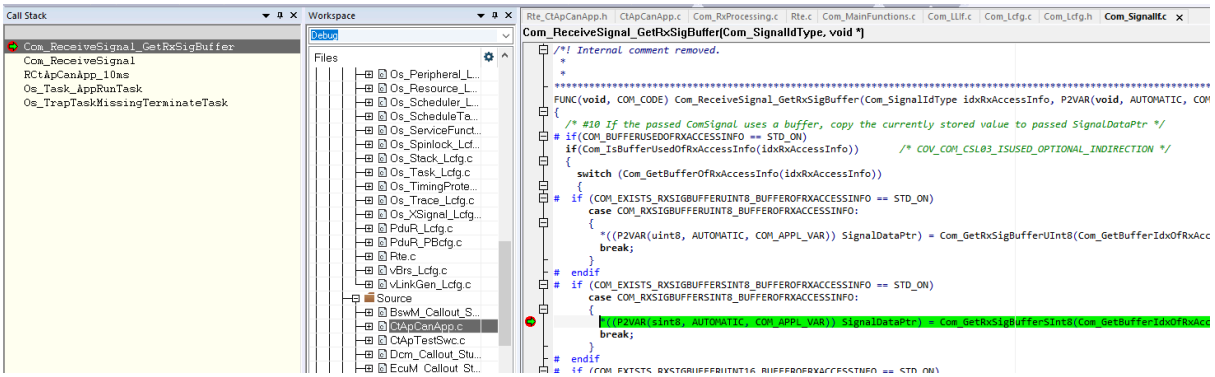




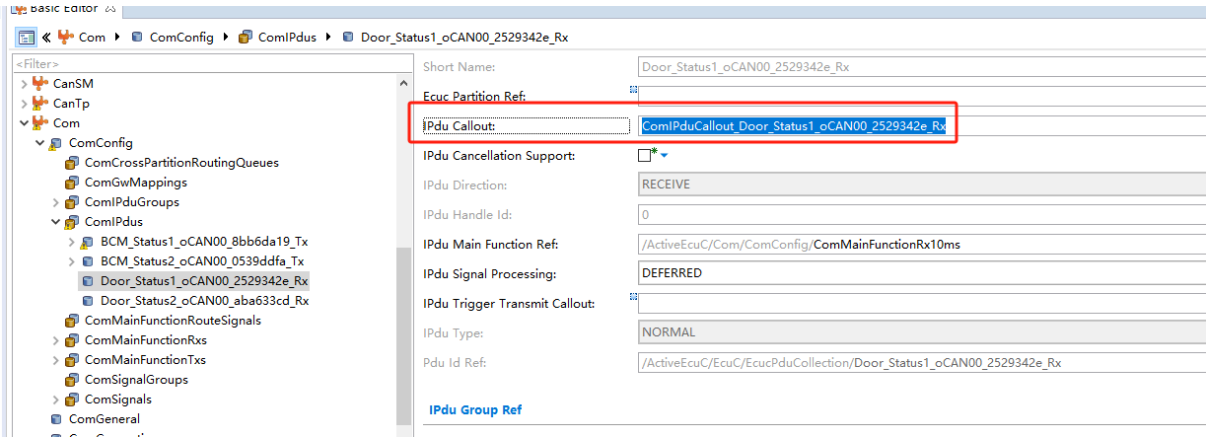
Com_RxSigBufferSInt8[2] 数据填充完成之后，接下来就等着等着应用调用Com_ReceiveSignal 去取对应信号的值了

2.2、Com_ReceiveSignal

应用调用Com_ReceiveSignal之后，相当于把Com_RxSigBufferSInt8[2] 对应的数据拷贝出去给应用，那么应用就取到对应数据了



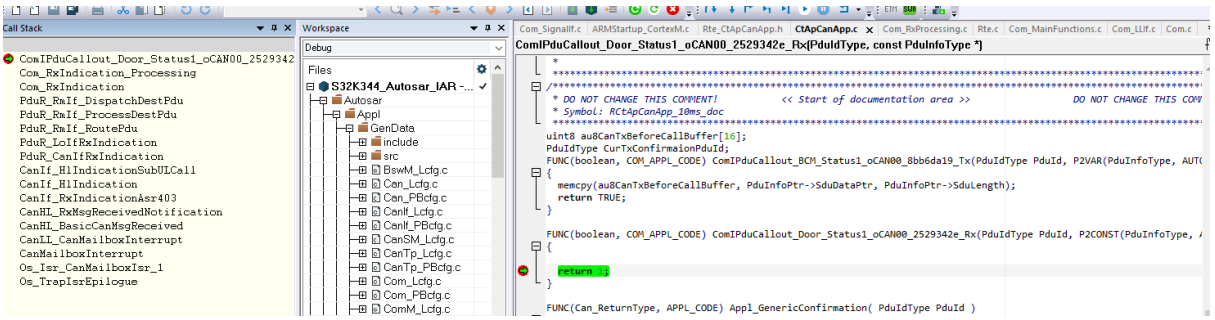
2.3、接收中断里的Rx ComIPduCallout 回调



然后用户实现接口

FUNC(boolean, COM_APPL_CODE) ComIPduCallout_Door_Status1_oCAN00_2529342e_Rx(PduIdType PduId, P2CONST(PduInfoType, AUTOMATIC, COM_APPL_DATA) PduInfoPtr)
{

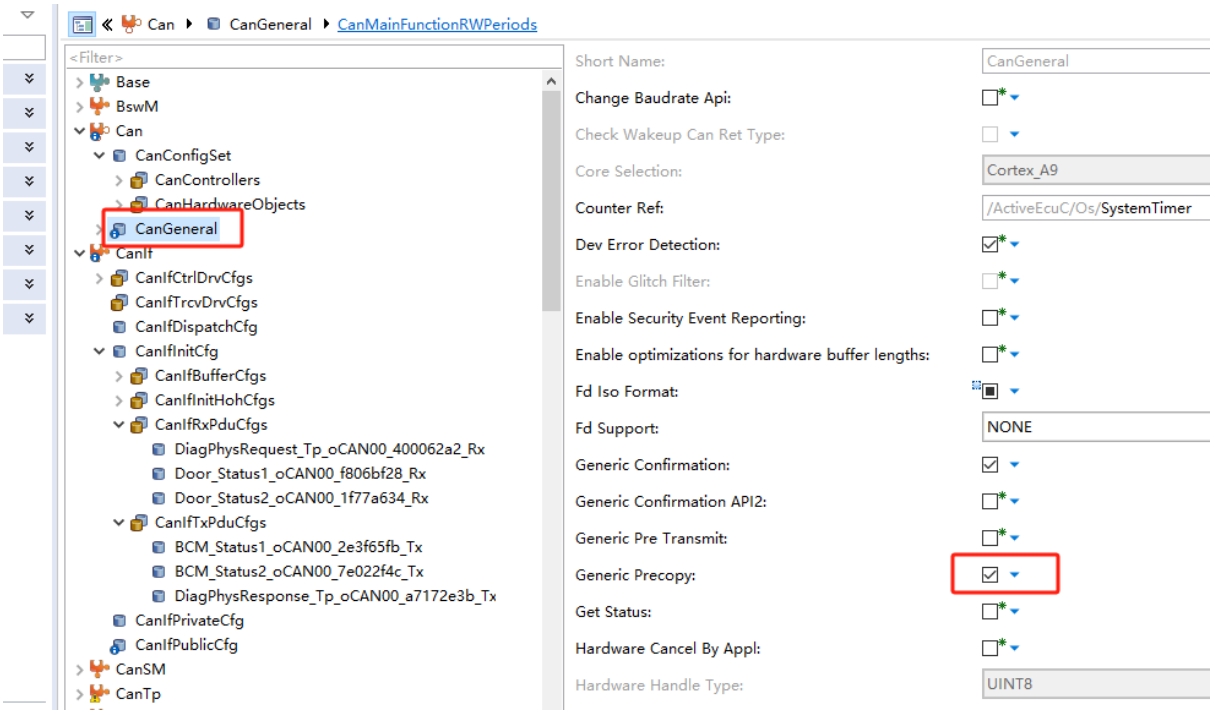
return TRUE;
}



可见Rx ComIPduCallout 是在中断里面完成的，当配置为中断接收模式时

2.4、CAN接收回调Appl_GenericPrecopy

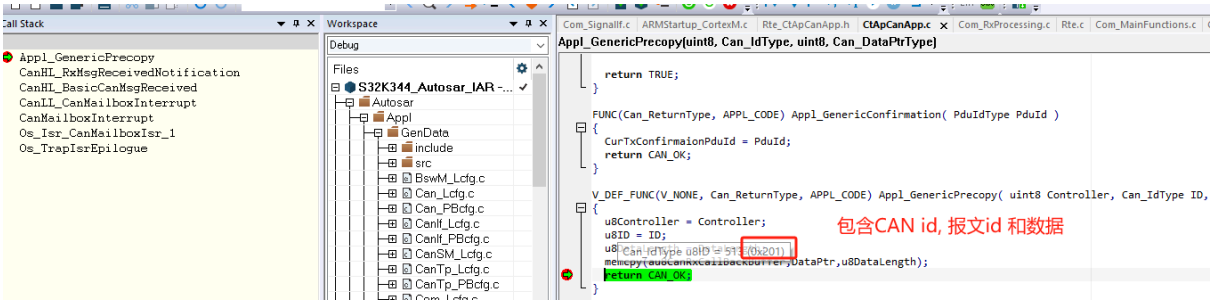
勾选Can的 Generic Precopy



然后用户实现接口

V_DEF_FUNC(V_NONE, Can_ReturnType, APPL_CODE) Appl_GenericPrecopy(uint8 Controller, Can_IdType ID, uint8 DataLength, Can_DataPtrType DataPtr)
{
u8Controller = Controller;
u8ID = ID;

```
u8DataLength = DataLength;
memcpy(au8CanRxCallBackBuffer,DataPtr,u8DataLength);
return CAN_OK;
}
```



从调用栈可见，Appl_GenericPrecopy 是在can中回调，可以处理所有can对应的报文，而Rx ComIPduCallout是在Com层的回调，针对的是某条报文